

Client Brief: AI Solutions for Energy, Recycling, and Environmental Monitoring

Consultancy: MindMesh Innovations

Engagement type: Multi-client AI development

Scope: Three independent projects delivered under a single engagement

Methods: MLP, CNN, RNN (deep learning)

Overview

MindMesh Innovations engaged a business analytics specialist to develop AI solutions for three of their clients, each facing a distinct prediction or classification challenge. The projects span tabular regression, image classification, and multivariate time-series forecasting, requiring different deep learning architectures tailored to each data type and business context.

Project 1: Solar Power Output Prediction

Client sector: Renewable energy / photovoltaic systems

Problem type: Regression (tabular data)

Background

Solar power generation is inherently variable. Output fluctuates with weather conditions, time of day, and panel orientation, making it difficult for grid operators and plant managers to forecast supply reliably. Inaccurate forecasts lead to energy waste, grid instability, and missed revenue from feed-in tariffs or spot markets.

Objective

Build and compare predictive models for solar power output using weather conditions and positional parameters (zenith and azimuth angles). The engagement requires developing both a linear regression baseline and multiple MLP architectures with varying depth and width, then identifying which model performs best and why.

Dataset

Tabular data containing solar power output measurements alongside weather variables and solar angle parameters. Training/test split: 70/30.

Deliverables

- Baseline linear regression model
 - Multiple MLP models with different architectures (varying hidden layers, node counts)
 - Performance comparison across all models with justification for the best-performing architecture
 - Assessment of deployment readiness for real-world forecasting
-

Project 2: Recyclable Material Classification from Images

Client sector: Waste management / recycling

Problem type: Multi-class image classification (6 classes)

Background

Effective recycling depends on accurate sorting of materials at processing facilities. Manual sorting is slow, inconsistent, and expensive. Automating classification through computer vision could improve throughput and reduce contamination rates in recycled material streams, directly supporting sustainability goals.

Objective

Develop CNN-based classifiers that can distinguish between six recyclable material types: cardboard, clothes, paper, metal, green glass, and white glass. The engagement requires experimenting with different CNN architectures (kernel sizes, layer depths, hidden node counts) to identify the best-performing configuration.

Dataset

2,691 real-world images across six material classes. Training/test split: 70/30.

Deliverables

- Multiple CNN architectures compared on classification performance
 - Identification of the worst-performing class with analysis of why it underperforms (e.g., visual similarity between classes, class imbalance, image quality)
 - Recommendations for improving classification accuracy through data collection, preprocessing, augmentation, or architectural changes
 - Discussion of how the solution could enhance real-world recycling operations
-

Project 3: Air Quality Forecasting (PM2.5 Concentration)

Client sector: Environmental protection / public health

Problem type: Multivariate time-series forecasting

Background

PM2.5 (particulate matter 2.5 micrometres or smaller) is one of the most harmful air pollutants, linked to respiratory disease, cardiovascular problems, and premature death. Accurate short-term forecasts of PM2.5 concentration are critical for issuing public health advisories, informing urban planning decisions, and shaping environmental policy.

Objective

Develop an RNN-based forecasting model for hourly PM2.5 concentration using a multivariate time-series dataset. The forecast horizon is 24 hours. The engagement requires experimenting with different RNN architectures and training configurations to maximise predictive accuracy.

Dataset

Hourly air quality measurements spanning March 2013 to February 2017, including PM2.5 concentration and related atmospheric indicators. Training set: all data before January 2016. Test set: January 2016.

Deliverables

- Multiple RNN architectures compared (e.g., vanilla RNN, LSTM, GRU with varying layer configurations)
 - Performance comparison with justification for the best model
 - Practical recommendations for improving forecasting capability in production
 - Discussion of how predictions can support public health responses and policy decisions
-

Cross-Cutting Considerations

- All three projects require rigorous experiment tracking: each architectural variant should be logged with its hyperparameters and performance metrics, not just the final model
- Models must be evaluated on held-out test sets with clearly defined split boundaries (no data leakage from future time periods in Project 3)
- Solutions should be assessed not just on test-set metrics but on practical deployment viability, including computational cost, retraining frequency, and interpretability for non-technical stakeholders
- Each project addresses a different data modality (tabular, image, sequential), so preprocessing pipelines and evaluation metrics will differ accordingly